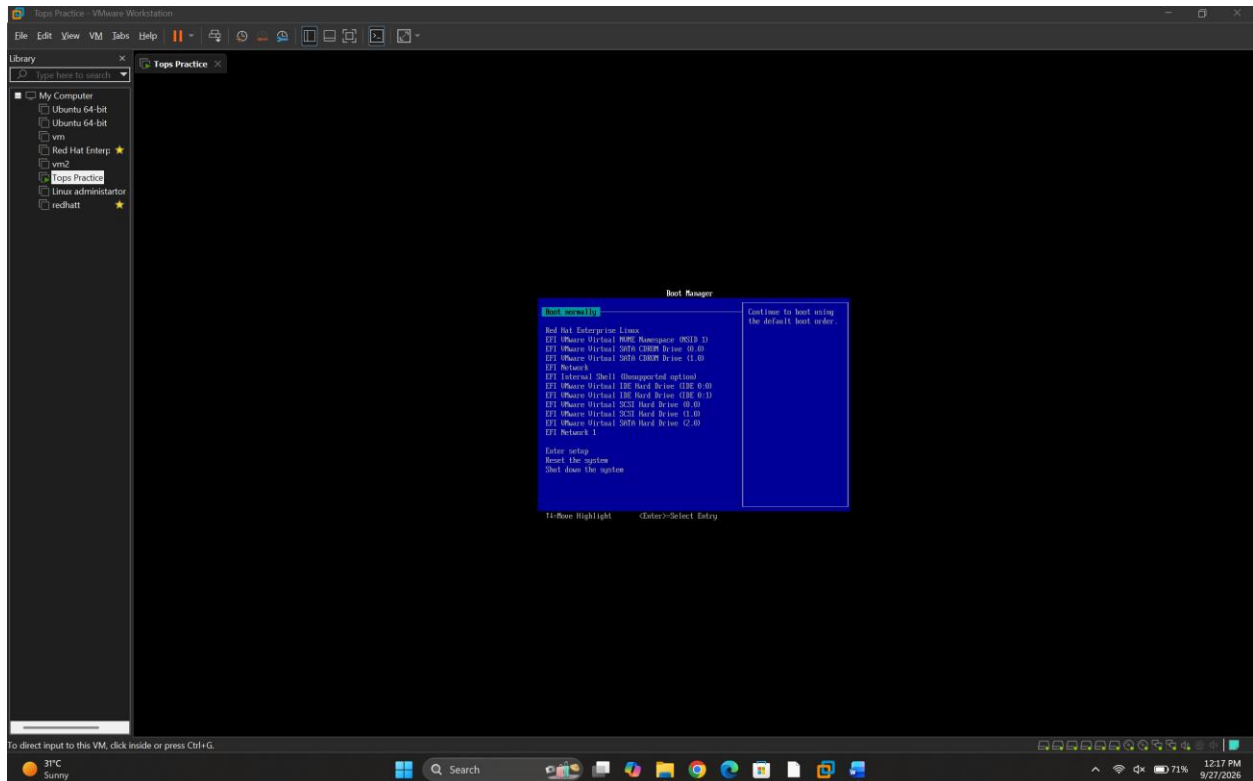


1.

Reboot your CentOS 7 or RHEL 8 virtual machine and, during startup, press the appropriate key to access the GRUB2 boot menu; take a screenshot of the GRUB menu screen.

 $\succ$ 

2.

Use the 'journalctl -b' command to view the system logs from the current boot and identify the timestamp when the multi-user.target was reached.

Hint: Search for 'Reached target Multi-User System' in the output.

 $\succ$

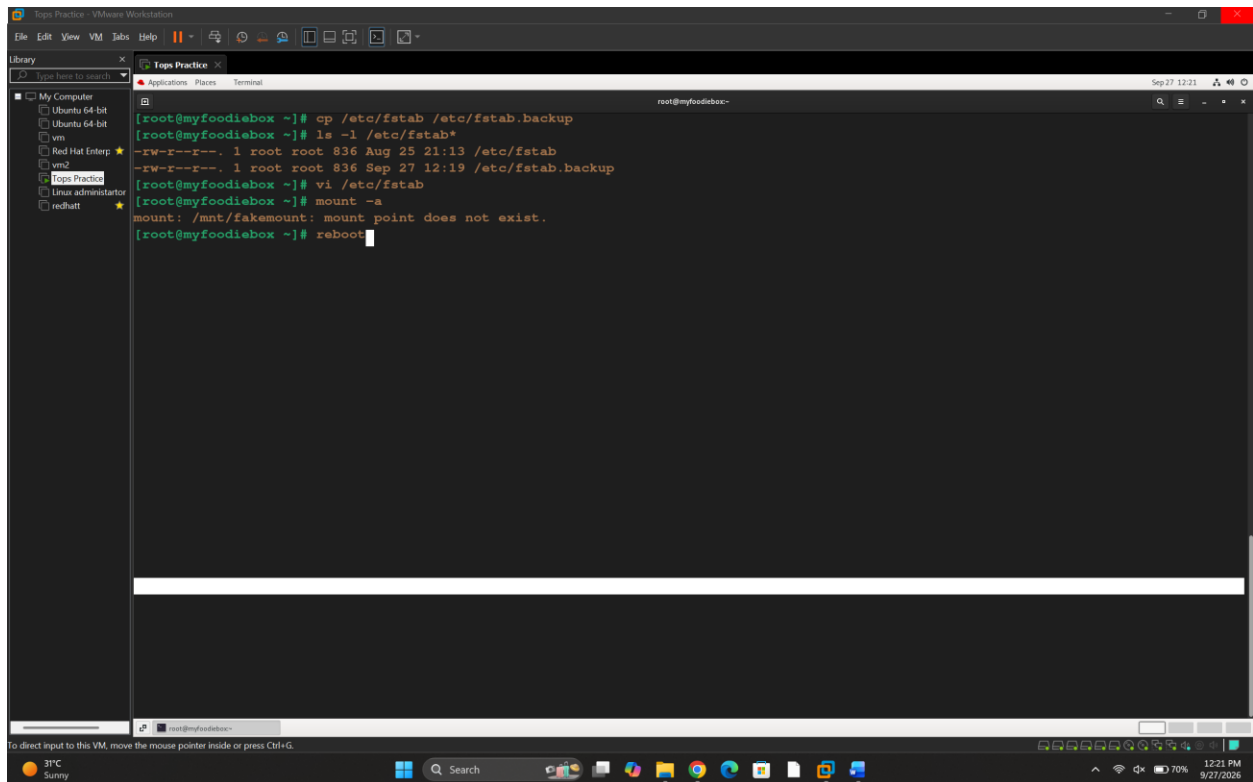
```
myfoodiebox systemd[1]: Reached target Multi-User System.
[root@myfoodiebox ~]# journalctl -b | grep -i "multi-user"
myfoodiebox systemd[1]: Reached target Multi-User System.
[root@myfoodiebox ~]# journalctl -b
Sep 27 12:17:23 localhost kernel: Linux version 5.14.0-70.13.1.el9_0.x86_64 (mockbuild@x86-vm-08.build.eng.bos.redhat.com) (gcc (GCC) 9.4.0) #1 SMP Tue Aug 11 22:03:12 PDT 2020; root=UUID=6d25fd57-0dfe-4506-b1d1-40707493c860
Sep 27 12:17:23 localhost kernel: The list of certified hardware and cloud instances for Red Hat Enterprise Linux 9 can be viewed at https://access.redhat.com/articles/6891461
Sep 27 12:17:23 localhost kernel: Command line: BOOT_IMAGE=(hdb,gpt2)/vmlinuz-5.14.0-70.13.1.el9_0.x86_64 root=UUID=6d25fd57-0dfe-4506-b1d1-40707493c860
Sep 27 12:17:23 localhost kernel: x86/fpu: Supporting XSAVE feature 0x001: 'x87 floating point registers'
Sep 27 12:17:23 localhost kernel: x86/fpu: Supporting XSAVE feature 0x002: 'SSE registers'
Sep 27 12:17:23 localhost kernel: x86/fpu: Supporting XSAVE feature 0x004: 'AVX registers'
Sep 27 12:17:23 localhost kernel: x86/fpu: xstate_offset[2]: 576, xstate_sizes[2]: 256
Sep 27 12:17:23 localhost kernel: x86/fpu: Enabled xstate features 0x7, context size is 832 bytes, using 'compacted' format.
Sep 27 12:17:23 localhost kernel: signal: max sigframe size: 1776
Sep 27 12:17:23 localhost kernel: BIOS-provided physical RAM map:
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] ACPI NVS
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000001000-0x0000000000001fff] reserved
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000002000-0x0000000000009fff] usable
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x000000000000c000-0x000000000000ffff] reserved
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000010000-0x000000000000e3bfff] usable
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x00000000000e03c000-0x00000000000e041fff] ACPI NVS
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x00000000000e042000-0x00000000000fb6efff] usable
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000fb6f000-0x0000000000fbdefff] reserved
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000fbdf000-0x0000000000fbfafff] ACPI data
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000fbfb000-0x0000000000fbfefff] ACPI NVS
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x0000000000fbff000-0x0000000007ffffff] usable
Sep 27 12:17:23 localhost kernel: BIOS-e820: [mem 0x00000000ffc00000-0x00000000ffc29fff] reserved
Sep 27 12:17:23 localhost kernel: NX (Execute Disable) protection: active
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d933018-0x0d93b057] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d933018-0x0d93b057] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d930018-0x0d932057] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d930018-0x0d932057] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d92d018-0x0d92f057] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d92d018-0x0d92f057] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d92b018-0x0d92c857] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d92b018-0x0d92c857] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d92b018-0x0d92ae57] usable ==> usable
Sep 27 12:17:23 localhost kernel: e820: update [mem 0x0d923018-0x0d92ae57] usable ==> usable
```

3.

Simulate a boot failure by editing the `/etc/fstab` file and adding a fake mount entry, then reboot your VM and document the error message that appears during boot.

Constraint: Restore your original `/etc/fstab` after testing to avoid permanent boot issues.

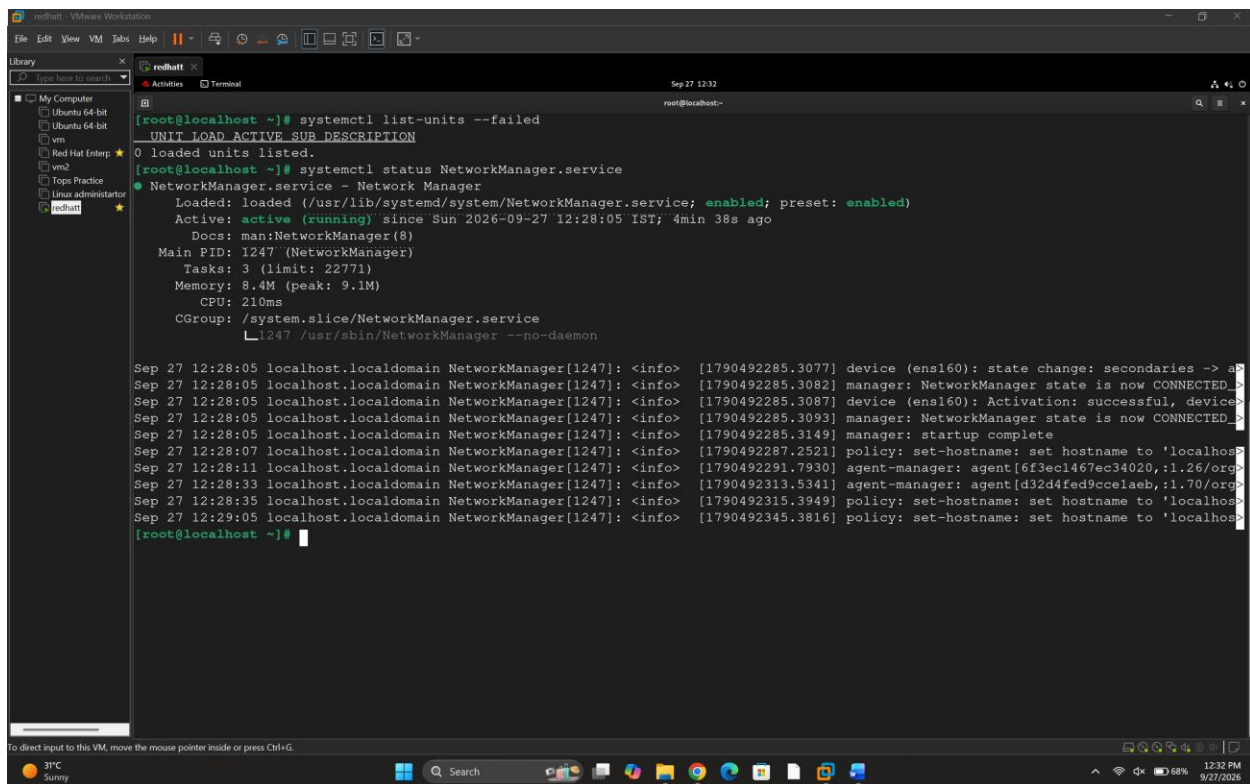
>



4.

Use the 'systemctl list-units --failed' command after booting to identify any failed services, and write down the name of one failed service and the command you would use to check its status.

>



```
[root@localhost ~]# systemctl list-units --failed
UNIT LOAD ACTIVE SUB DESCRIPTION
0 loaded units listed.

[root@localhost ~]# systemctl status NetworkManager.service
● NetworkManager.service - Network Manager
   Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-09-27 12:28:05 IST; 4min 38s ago
     Docs: man:NetworkManager(8)
   Main PID: 1247 (NetworkManager)
    Tasks: 3 (limit: 22771)
   Memory: 8.4M (peak: 9.1M)
      CPU: 210ms
   CGroup: /system.slice/NetworkManager.service
           └─1247 /usr/sbin/NetworkManager --no-daemon

Sep 27 12:28:05 localhost.localdomain NetworkManager[1247]: <info> [1790492285.3077] device (ens160): state change: secondaries -> a
Sep 27 12:28:05 localhost.localdomain NetworkManager[1247]: <info> [1790492285.3082] manager: NetworkManager state is now CONNECTED
Sep 27 12:28:05 localhost.localdomain NetworkManager[1247]: <info> [1790492285.3087] device (ens160): Activation: successful, device
Sep 27 12:28:05 localhost.localdomain NetworkManager[1247]: <info> [1790492285.3093] manager: NetworkManager state is now CONNECTED
Sep 27 12:28:05 localhost.localdomain NetworkManager[1247]: <info> [1790492285.3149] manager: startup complete
Sep 27 12:28:07 localhost.localdomain NetworkManager[1247]: <info> [1790492287.2521] policy: set-hostname: set hostname to 'localhos
Sep 27 12:28:11 localhost.localdomain NetworkManager[1247]: <info> [1790492291.7930] agent-manager: agent[6f3ec1467ec34020;1.26/org
Sep 27 12:28:33 localhost.localdomain NetworkManager[1247]: <info> [1790492313.5341] agent-manager: agent[d32d4fed9ccelaeb;1.70/org
Sep 27 12:28:35 localhost.localdomain NetworkManager[1247]: <info> [1790492315.3949] policy: set-hostname: set hostname to 'localhos
Sep 27 12:29:05 localhost.localdomain NetworkManager[1247]: <info> [1790492345.3816] policy: set-hostname: set hostname to 'localhos
[root@localhost ~]#
```

5.

Suppose your system is stuck at the emergency shell prompt during boot. Use ChatGPT to generate a step-by-step recovery plan for this scenario, then try following those steps on your VM and note any differences or extra steps you had to take.

>

```
redhat - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
  Ubuntu 64-bit
  Ubuntu 64-bit
  vm
  Red Hat Enterp...
  vm2
  Tops Practice
  Linux administrator
  redhat
Sep 27 12:35
root@localhost~
[root@localhost ~]# systemctl --failed
UNIT LOAD ACTIVE SUB DESCRIPTION
0 loaded units listed.
[root@localhost ~]# journalctl -xb
Sep 27 12:27:55 localhost kernel: Linux version 5.14.0-687.46.1.el9_8.x86_64 (mockbuild@x86-64-04.build.eng.rdu2.redhat.com) (gcc (GCC)
Sep 27 12:27:55 localhost kernel: The list of certified hardware and cloud instances for Red Hat Enterprise Linux 9 can be viewed at
Sep 27 12:27:55 localhost kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-5.14.0-687.46.1.el9_8.x86_64 root=/dev/mapper/rhel-root
Sep 27 12:27:55 localhost kernel: BIOS-provided physical RAM map:
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] ACPI NVS
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x0000000000001000-0x0000000000001fff] reserved
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x0000000000002000-0x0000000000009fff] usable
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x000000000000c000-0x000000000000ffff] reserved
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x0000000000010000-0x000000000000dab3fff] usable
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000dab4000-0x00000000000dab5fff] ACPI data
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000dab6000-0x00000000000dabcbfff] usable
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000dacc000-0x00000000000daccfff] ACPI data
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000dacc000-0x00000000000e03bfff] usable
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000e03c000-0x00000000000e041fff] ACPI NVS
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000e042000-0x00000000000fb6efff] usable
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000fb6f000-0x00000000000fbdefff] reserved
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000fbdf000-0x00000000000fbfafff] ACPI data
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000fbfb000-0x00000000000fbfefff] ACPI NVS
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000fbff000-0x00000000000bffffff] usable
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x00000000000ffc0000-0x00000000000ffc29fff] reserved
Sep 27 12:27:55 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000013fffff] usable
Sep 27 12:27:55 localhost kernel: NX (Execute Disable) protection: active
Sep 27 12:27:55 localhost kernel: APIC: Static calls initialized
Sep 27 12:27:55 localhost kernel: e820: update [mem 0x0dac2018-0x0dac057] usable ==> usable
Sep 27 12:27:55 localhost kernel: e820: update [mem 0x0daba018-0x0dabc057] usable ==> usable
Sep 27 12:27:55 localhost kernel: e820: update [mem 0x0dab8018-0x0dab9857] usable ==> usable
Sep 27 12:27:55 localhost kernel: e820: update [mem 0x0dab6018-0x0dab7e57] usable ==> usable
Sep 27 12:27:55 localhost kernel: extended physical RAM map:
Sep 27 12:27:55 localhost kernel: reserve setup_data: [mem 0x0000000000000000-0x0000000000000fff] ACPI NVS
Sep 27 12:27:55 localhost kernel: reserve setup_data: [mem 0x00000000000001000-0x00000000000001fff] reserved
Sep 27 12:27:55 localhost kernel: reserve setup_data: [mem 0x00000000000002000-0x00000000000009ffff] usable
To direct input to this VM, move the mouse pointer inside or press Ctrl+G
```

```
redhat - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
  Ubuntu 64-bit
  Ubuntu 64-bit
  vm
  Red Hat Enterp...
  vm2
  Tops Practice
  Linux administrator
  redhat
Sep 27 12:35
root@localhost~
n is closed
[root@localhost ~]# cat /etc/fstab
#
# /etc/fstab
# Created by anaconda on Sun Sep 13 05:40:52 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rhel-root / xfs defaults 0 0
UUID=e6ba69dc-9646-427e-b3db-0c7d1e0b53a9 /boot xfs defaults 0 0
UUID=75FE-EE1C /boot/efi vfat umask=0077,shortname=winnt 0 2
/dev/mapper/rhel-home /home xfs defaults 0 0
/dev/mapper/rhel-var /var xfs defaults 0 0
/dev/mapper/rhel-swap none swap defaults 0 0
[root@localhost ~]# lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sr0 11:0 1 90.8M 0 rom /run/media/redhat/CDROM
sr1 11:1 1 8G 0 rom /run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1 259:0 0 50G 0 disk
├─nvme0n1p1 259:1 0 2G 0 part /boot/efi
├─nvme0n1p2 259:2 0 4G 0 part /boot
├─nvme0n1p3 259:3 0 44G 0 part
└─┬─rhel-root 253:0 0 30G 0 lvm /
   └─┬─rhel-swap 253:1 0 200M 0 lvm [SWAP]
     └─┬─rhel-var 253:2 0 10G 0 lvm /var
       └─rhel-home 253:3 0 2G 0 lvm /home
[root@localhost ~]# systemctl daemon-reload
[root@localhost ~]# systemctl default
[root@localhost ~]# systemctl status
● localhost.localdomain
  State: running
  Units: 400 loaded (incl. loaded aliases)
  Jobs: 0 queued
To direct input to this VM, move the mouse pointer inside or press Ctrl+G
```

redhat - VMware Workstation

File Edit View VM Help

Library

My Computer

- Ubuntu 64-bit
- Ubuntu 64-bit
- vm
- Red Hat Enterp. ★
- vm2
- Tools Practice
- Linux administrator
- redhat ★

Activities Terminal

Sep 27 12:35

root@localhost~

```
_system.slice
├─ModemManager.service
│   └─1029 /usr/sbin/ModemManager
├─NetworkManager.service
│   └─1247 /usr/sbin/NetworkManager --no-daemon
├─accounts-daemon.service
│   └─960 /usr/libexec/accounts-daemon
├─atd.service
│   └─1303 /usr/sbin/atd -f
├─auditd.service
│   └─909 /sbin/auditd
│   └─911 /usr/sbin/sedispach
├─avahi-daemon.service
│   └─948 "avahi-daemon: running [linux.local]"
│   └─974 "avahi-daemon: chroot helper"
├─chronyd.service
│   └─992 /usr/sbin/chronyd -F 2
├─crond.service
│   └─1304 /usr/sbin/crond -n
├─cups.service
│   └─1260 /usr/sbin/cupsd -l
├─dbus-broker.service
│   └─942 /usr/bin/dbus-broker-launch --scope system --audit
│   └─947 dbus-broker --log 4 --controller 9 --machine-id 3a404e739c9246a199ce15f22fc7effa --max-bytes 536870912 --max-fds 2
├─firewalld.service
│   └─1030 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid
└─[root@localhost ~]# journalctl -b -p err
Sep 27 12:28:00 localhost kernel: piix4_smbus 0000:00:07.3: SMBus base address uninitialized - upgrade BIOS or use force_addr=0xaddr
Sep 27 12:28:02 localhost alsactl[999]: alsa-lib main.c:1804: (snd_use_case_mgr_open) [error.ucm] failed to import hw:0 use case confi
Sep 27 12:28:05 localhost.localdomain smbd[1288]: [2026/09/27 12:28:05.702738, 0] ../../lib/param/loadparm.c:750(lpcfg_map_parameter)
Sep 27 12:28:05 localhost.localdomain smbd[1288]: Unknown parameter encountered: "user list"
Sep 27 12:28:05 localhost.localdomain smbd[1288]: [2026/09/27 12:28:05.703326, 0] ../../lib/param/loadparm.c:2044(lpcfg_do_service_p
Sep 27 12:28:05 localhost.localdomain smbd[1288]: Ignoring unknown parameter "user list"
Sep 27 12:28:28 localhost.localdomain gdm-password[2806]: gkr-pam: unable to locate daemon control file
Sep 27 12:28:36 localhost.localdomain gdm-wayland-session[1557]: GLib: Source ID 2 was not found when attempting to remove it
Sep 27 12:28:36 localhost.localdomain gdm-launch-environment[1343]: GLib-GObject: g_object_unref: assertion 'G_IS_OBJECT (object)' failed
[root@localhost ~]#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G

31°C Sunny

Search

12:35 PM 9/27/2026